

RIONEGRO, Colombia

WMO: 801120

Lat: 6.165N

Lon: 75.423W

Elev: 7028

StdP: 11.33

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	50.4	51.9	46.3	60.5	68.2	48.1	64.7	66.8	15.1	68.4	13.6	69.8	0.4	240	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.3	75.2	60.5	73.8	60.1	73.5	60.0	63.4	70.2	62.8	69.7	62.2	68.9	11.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	105.4	64.4	61.0	104.6	64.1	60.7	103.2	63.7	33.0	70.3	32.2	69.4	31.8	68.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.5	12.0	DB	45.8	79.9	3.6	3.3	43.2	82.3	41.1	84.3	39.0	86.1	36.4	88.5	(4)
(5)			WB	45.8	67.8	1.9	3.7	44.5	70.5	43.4	72.6	42.3	74.7	40.9	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	62.8	63.4	63.5	63.8	64.2	63.7	63.3	63.4	63.6	63.1	63.0	63.0	(6)	
	DBStd	1.74	1.86	2.00	1.89	1.71	1.71	1.52	1.47	1.54	1.49	1.75	1.65	1.72	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	644	71	52	54	43	35	44	55	53	47	62	62	66	(9)	
	CDD50	4887	397	374	419	414	440	410	412	415	407	407	390	402	(10)	
	CDD65	52	3	6	7	7	10	4	2	3	3	3	2	2	(11)	
	CDH74	100	8	11	12	10	17	7	5	8	11	5	2	4	(12)	
	CDH80	1	0	0	0	0	1	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.3	3.3	3.1	2.9	3.2	3.9	4.2	4.2	3.7	3.0	2.8	3.0	(14)	
(15) Precipitation	PrecAvg	76.1	2.0	2.8	4.0	7.8	9.0	7.0	6.4	8.2	8.7	9.8	6.9	3.4	(15)	
	PrecMax	93.1	4.3	6.7	8.9	12.8	14.8	11.5	11.2	13.7	12.8	14.6	13.7	6.7	(16)	
	PrecMin	64.4	0.3	0.5	1.6	4.2	3.1	3.4	1.0	3.3	4.5	6.1	2.9	0.9	(17)	
	PrecStd	7.5	1.1	1.5	1.7	2.4	2.6	2.0	2.4	2.7	2.3	2.1	2.5	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	75.4	75.5	75.3	75.5	75.2	74.8	75.2	75.3	74.6	73.6	73.8	(19)	
		MCWB	60.2	60.6	61.3	61.5	61.3	60.6	59.7	59.6	59.5	60.0	60.5	59.9	(20)	
	2%	DB	73.2	73.5	73.6	73.5	73.7	73.6	73.5	73.6	73.7	73.1	71.9	72.3	(21)	
		MCWB	60.0	60.0	60.8	61.1	61.0	60.1	59.5	59.4	59.5	59.9	60.2	59.8	(22)	
	5%	DB	71.5	71.8	71.9	71.9	72.9	72.9	72.7	73.0	73.0	71.5	70.9	71.3	(23)	
		MCWB	59.6	59.9	60.7	61.1	60.9	60.0	59.2	59.3	59.4	59.8	60.2	59.7	(24)	
	10%	DB	69.8	70.1	70.1	70.1	71.3	71.5	71.6	71.6	71.5	69.8	69.4	69.7	(25)	
		MCWB	59.4	59.8	60.5	61.0	60.9	59.7	58.8	59.0	59.3	59.7	60.1	59.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	63.6	63.8	63.9	63.9	63.3	62.7	62.9	62.9	62.9	62.9	62.9	(27)	
		MCDB	70.4	70.1	70.2	69.9	71.0	70.2	70.5	70.7	70.2	70.0	69.3	69.3	(28)	
	2%	WB	62.0	62.3	62.6	62.9	62.8	62.1	61.2	61.5	61.8	61.7	61.9	61.8	(29)	
		MCDB	68.6	69.1	69.3	69.4	69.8	69.3	68.7	68.9	69.0	68.4	68.0	68.1	(30)	
	5%	WB	61.1	61.6	61.9	62.2	62.1	61.2	60.7	60.9	61.0	61.0	61.2	61.1	(31)	
		MCDB	67.3	67.8	68.2	68.5	68.8	68.2	68.0	67.9	68.0	67.3	67.1	67.2	(32)	
	10%	WB	60.4	60.9	61.1	61.5	61.4	60.8	60.1	60.2	60.3	60.3	60.6	60.4	(33)	
		MCDB	66.1	66.6	67.2	67.5	67.8	67.5	67.3	67.3	67.1	66.2	66.3	66.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	15.6	15.2	14.9	15.3	16.7	17.3	17.6	16.9	15.2	14.5	14.9	(35)	
		MCDBR	17.9	17.7	17.3	16.8	17.4	18.5	18.9	19.2	18.8	17.1	16.2	16.9	(36)	
	5% WB	MCWBR	7.0	6.8	6.8	6.7	6.6	7.0	6.9	7.1	6.8	6.6	6.2	6.5	(37)	
		MCWBR	15.7	15.9	15.7	15.5	15.8	16.5	17.2	17.1	16.9	15.5	14.7	15.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.356	0.374	0.404	0.380	0.346	0.333	0.340	0.349	0.359	0.363	0.342	0.347	(40)	
		taud	2.475	2.414	2.323	2.407	2.537	2.576	2.536	2.485	2.455	2.470	2.557	2.529	(41)	
	Ebn at Noon	Ebn at Noon	305	301	292	293	298	298	297	298	300	301	307	306	(42)	
		Edh at Noon	36	39	43	39	33	31	33	35	37	36	32	33	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1496	1507	1428	1425	1521	1618	1694	1709	1648	1472	1388	1418	(44)	
	RadStd	RadStd	101	126	92	62	66	81	94	99	108	67	71	86	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.43	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+0.43	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page